

Research Methodology and Literature Review

Science

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Research Methodology and Literature Review (A11243)

Short Title: Res Methodology & Lit Review
Faculty Science and Computing
Department Science
Cluster Placement and Projects
Author KGRENNAN
Credits 5

Level: Advanced

Description of Module / Aims

This module will develop techniques that are necessary in a research environment, such as accessing, referencing and utilising public literature resources to prepare a literature review of a specialised research topic. In addition, the module will develop the skills required to design a project management plan to implement a practical project related to this research topic.

Programmes

RESA-0089	BSc (Hons) in Applied Biology with Quality Management (WD_SQMBI_B)	4 / 1 / M
RESA-0089	BSc (Hons) in Applied Chemistry with Quality Management (WD_SQMCH_B)	4 / 1 / M
RESA-0089	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	4 / 7 / M
RESA-0089	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	4 / 7 / M
RESA-0089	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	4 / 7 / M

Indicative Content

- Search strategies and tools for searching the scientific literature, using online databases, books, journals and Internet resources
- Referencing the literature using the Harvard and Vancouver systems & dedicated reference manager software
- Critical analysis and evaluation of the literature related to a specialised research topic
- Formatting and design of a literature review document using the built-in tools of MS Word
- Formatting tables, graphs, schemes and images to a high standard for a research document
- Advanced scientific writing skills
- Oral presentation of literature review
- Design and present a project management plan and Chemical Safety Forms to implement a practical project on a specialised research topic

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Plan a literature search to search the scientific literature and databases relevant to a specialised research topic.
2. Compile a review of the scientific literature information on a specialised research topic using the built-in tools of MS Word.
3. Evaluate scientific literature relevant to a specialised research topic.
4. Manage referencing resources to support a literature review.
5. Produce an oral presentation of the research findings.
6. Design a project management plan for a practical project related to a specialised research topic.

Learning and Teaching Methods

- IT tutorials.
- Library training.
- Consultations with project supervisor.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Project</i>	70%	1,2,3,4
<i>Presentation</i>	20%	5
<i>Assignment</i>	10%	6

Assessment Criteria

- <40%:** Inadequate understanding of the fundamentals of research methodologies and insufficient competence in their use. Insufficient level of knowledge of chosen research topic.
- 40%-49%:** Ability to show a basic knowledge of research methodologies and basic competency in their use. Basic level of knowledge of chosen research topic.
- 50%-59%:** As well as the above, the student will be able to demonstrate an understanding of some of the complexities of research methodologies and a good competency in their use. Display an adequate level of knowledge of chosen research topic supplemented by an ability to present this knowledge in a clear and concise manner.
- 60%-69%:** In addition to the above, the student will demonstrate a more detailed knowledge of research methodologies and a high level of competence in their use. Display a high level of knowledge of chosen research topic, supplemented by an ability to present this knowledge to a high technical standard.
- 70%-100%:** All of the above to an excellent level. In addition, the student will demonstrate an advanced knowledge of research methodologies, as well as excellent and proficient competence in their use. Display a thorough level of knowledge of chosen research topic, supplemented by an ability to critically evaluate and discuss the topic in relation to a wider scientific context. The above advanced knowledge of the topic will be supported by superior reporting and presentation skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial	24	
Other	12	
Independent Learning	99	

Essential Material

- "Research Methodology & Literature Review Moodle page." <http://moodle.wit.ie>
- "WIT library catalogue." <http://library.wit.ie/>
- "WIT library databases." <http://library.wit.ie/Resources/databases>

Supplementary Material

- Lebrun, J.L. _Scientific writing 2.0: a reader's and writer's guide_. Hackensack, NJ: New World Scientific, 2011.
- Marder , M.P. _Research methods for science_. Cambridge: Cambridge University Press, 2011.
- Neville, C. _Complete guide to referencing and avoiding plagiarism_. Maidenhead , NY: Open University, 2010.

Requested Resources

- Room Type: Computer Lab